



Technical Appendix

Dependence Without Predictability in Forecasting Seven African Great Lakes from Fused Satellite Altimetry

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Code Repository Link: <https://github.com/SharpRS03/UoM-ERP-BM-10909914>

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1. Purpose and Scope

This document and the repository it links to are the additional materials for the extended research project. Together they should be able to be used to replicate the results of this study. The repository holds the code as it was executed, and reference copies of key results. This document has everything that is not code, where the source data comes from and how it was requested, what software and hardware produced the results, the order the stages run in, every parameter setting. Following the guidance the source data itself is not redistributed, all of it is public and each product is identified in this document by provider, version and access route.

1.1. What is and is not redistributed

Following the guidance, the source data is not redistributed. All of it is public and each product is identified in Section 3 by provider, version and access route, with the exact request parameters where necessary. There are two deliberate exceptions, explained below:

- The two climate index series `dmi.csv` and `nino34.csv`, a few kilobytes each. These are committed because the Dipole Mode Index is revised at source and the reported lags rely on the exact version cited at that specific time.
- The DAHITI water-level records for five lakes, which are embedded as literal values inside the collation notebooks as opposed to being read from a file. They were transcribed during development, and the code has been left as it ran; a reproducer therefore needs no DAHITI account and no DAHITI download.

Separately from the source data, the repository commits a set of reference outputs. These are the author's own results, not inputs to the analysis, and they are supporting material to be used so a reproducer can compare their own run against the numbers the report quotes. Therefore, the reproducer can tell whether a difference arose in the data, in their environment, or in their execution of the pipeline. Section 2.3 outlines what exactly makes up the reference material.

2. The Repository

2.1. Structure

Within the repository are 23 .ipynb files, three .py files, three folders for data from Copernicus, G-Realm, and the Climate Indices, and two additional folders, one for Reference Outputs which are described in 2.3, and one for documents called docs containing DATA.md and this appendix as a pdf. The Code Outputs folder is created upon the running of the first notebook Stage 00. The .ipynb files read the 4 data files, and the ones that require it read the .py files and produce their respective outputs in the Code Outputs folder.

2.2. What each part is for

The .ipynb files are the analysis, twenty-three stages in numerical order. The .py files are three modules imported by stages 10, 16, and 17. The G-Realm and Copernicus folders are empty in the repository but are designed to store the various parts of lake data with Copernicus also holding the climate data, with the DAHITI data being committed within the code. The Climate Indices folder holds both DMI and Niño 3.4 data. Reference Outputs holds the author's own copies of a few key results, in the same subfolder layout a run writes into Code Outputs, the pipeline does not read Reference Outputs.

2.3. The reference outputs, and what was intentionally excluded

Throughout the repository the only work included is work a reported result depends on, the stages that produce it, the exploratory stages that fixed the specification, and the reference output needed to verify it. Superseded notebooks and artefacts made for progress-meetings and the video presentation are absent from the repository. Reference Outputs/ contains two things:

- Every file named in the provenance map of Section 7, that is the file behind each numbered figure and table in the report.
- The four derived data products of Section 3.6, which are also the entry point for a partial reproduction.

A full run of the pipeline would write considerably more than this into Code Outputs, intermediate diagnostics, per-variant adjacency matrices, comparison figures superseded by the report's own. None of that has been committed, but can all be reproduced. Alongside those mentioned above, Reference Outputs also carries the handful of intermediate files that a later stage reads as an input: the adjacency matrices, the lead-lag table, the shared baseline predictions, and the selected orders. This is so that a partial reproduction beginning part-way through the pipeline has everything it needs.

3. Data and access

Seven lakes of the East African Rift are studied: Victoria, Tanganyika, Turkana, Malawi, Albert, Edward and Kivu. Lakes were selected on sustained altimetric coverage across the satellite era, which excludes smaller Rift lakes whose surface area falls below the reliable retrieval threshold.

3.1. Sources

Product	Provider	Identifier	Version	Accessed
Lake water levels from 1992 to present derived from satellite observations	Copernicus Climate Change Service	doi:10.24381/cds.5714c668	v5.0	26 Feb 2026
Global Reservoir and Lake Monitor (G-REALM)	USDA Foreign Agricultural Service/ NASA GSFC	https://earth.gsfc.nasa.gov/gwm/lake/Index	Per-lake target IDs, Table T 2	27 Feb 2026
Database for Hydrological Time Series of Inland Waters (DAHITI)	DGFI, Technical University of Munich	https://dahiti.dgfi.tum.de/en/	Software v8.0	29 Feb 2026
ERA5 post-processed daily statistics on single levels from 1940 to present	Copernicus Climate Data Store	https://doi.org/10.24381/cds.4991cf48	v1.0	19 Mar 2026
Dipole Mode Index, monthly	NOAA Physical Sciences Laboratory	https://psl.noaa.gov/data/timeseries/month/DMI/	HadISST 1.1 input	06 August 2026
Niño 3.4 SST index, monthly	NOAA Physical Sciences Laboratory	https://psl.noaa.gov/data/timeseries/month/Nino34_CPC/	NOAA ERSST v5	20 June 2026

TABLE T 1: Source Products. Per-lake identifiers and missions are in Table T 2; record extents in Table T 3.

DAHITI asks to be cited by this methodological paper: Schwatke, C., Dettmering, D., Bosch, W. and Seitz, F. (2015) DAHITI — an innovative approach for estimating water level time series over inland waters using multi-mission satellite altimetry. *Hydrology and Earth System Sciences*, 19, pp. 4345–4364. doi:10.5194/hess-19-4345-2015.

The two Copernicus products carry a shared attribution requirement: cite the CDS catalogue entry, and give clear and visible attribution to the Copernicus programme for each product used.

Copernicus Climate Change Service, Climate Data Store, (2020): Lake water levels from 1992 to present derived from satellite observation. Copernicus Climate Change Service (C3S) Climate Data Store (CDS). DOI: [10.24381/cds.5714c668](https://doi.org/10.24381/cds.5714c668) (Accessed on 26-Feb-2026)

This work contains modified Copernicus Climate Change Service information (2026); neither the European Commission nor ECMWF is responsible for any use that may be made of the Copernicus information or data it contains.

The Dipole Mode Index is calculated at NOAA PSL from the HadISST 1.1 dataset and is cited to *Possible impacts of Indian Ocean Dipole mode events on global climate. Authors: Saji, N.H. and Yamagata, T, CLIMATE RES, 25 (2): 151-169 DEC 5 2003*. The Niño 3.4 series is derived from NOAA ERSST v5 and has no suggested citation of its own.

3.2. Per-lake altimetry identifiers

Lake	G-REALM ID	Cadence	DAHITI ID	Missions contributing
Victoria	000314	10-day	2	TOPEX, Poseidon, Jason-1/2/3, Sentinel-6A
Tanganyika	000315	10-day	25	TOPEX, Poseidon, Jason-1/2/3, Sentinel-6A
Malawi	000317	10-day	-	TOPEX, Poseidon, Jason-1/2/3, Sentinel-6A (G-REALM only)
Turkana	000093	10-day	-	TOPEX, Poseidon, Jason-1/2/3, Sentinel-6A (G-REALM only)
Albert	000405	Monthly	85	Sentinel-3A; DAHITI add ICESat-2 (ATL13v7, beams 1L-3R), Jason-1 (SGDR-E), Saral (SGDR-F), Sentinel-3A/3B (NTC) Topex/Poseidon (Side-B, GDR-F)
Edward	000004	Monthly	86	Sentinel-3A; DAHITI adds SARAL and Envisat v3
Kivu	000069	Monthly	87	Sentinel-3A; DAHITI adds SARAL and Envisat v3

TABLE T 2: Per-lake altimetry target identifiers and contributing missions

3.3. Per-lake records

Lake	Repositories	Months	Start	End	Mean (m)	SD (m)	Range (m)
Albert	C3S, G-REALM, DAHITI	370	1995-06	2026-03	619.92	1.026	4.34
Edward	C3S, G-REALM, DAHITI	371	1995-05	2026-03	914.04	0.349	1.60
Kivu	C3S, G-REALM, DAHITI	371	1995-05	2026-03	1461.39	0.384	1.79
Malawi	C3S, G-REALM	403	1992-09	2026-03	474.16	0.738	3.88
Tanganyika	C3S, G-REALM, DAHITI	401	1992-10	2026-02	769.56	0.805	4.13
Turkana	C3S, G-REALM	402	1992-10	2026-03	363.51	2.067	9.06
Victoria	C3S, G-REALM, DAHITI	402	1992-09	2026-02	1135.37	0.638	3.02

TABLE T 3: Full-record extent and descriptive starts per lake, after fusion.

3.4. ERA5 Specific Request

Retrieved from the Copernicus Climate Data Store. This is a single regional download; per-lake series are produced afterwards by cropping to per-lake bounding boxes inside it and averaging over the box.

Field	Value
product_type	reanalysis
variable	2m_temperature, total_precipitation, potential_evaporation
year	1992-2026
month	All
day	All
daily_statistic	daily_mean for 2m_temperature; daily_sum for total_precipitation and potential_evaporation
time_zone	utc+00:00
frequency	1_hourly
area (N/W/S/E)	6, 27, -15, 38
data_format	netcdf

TABLE T 4: ERA5 request, reproduced from notebook 04. Temperature is retrieved one year per file; precipitation and potential evaporation are retrieved in half-year halves, because a full year of two accumulated variables exceeds the CDS request limit.

Part of the archive was retrieved manually before the download was automated, so Notebook 04 covers 2m temp from 2015 and precipitation and potential evaporation from 2021; the earlier years were requested through the CDS website, using the same dataset and variables. Precipitation and potential evaporation are read from half-year files, so a missing half does not flag as an error. To avoid this causing issues, confirm the output Lake_Climate_Monthly.xlsx has 400 rows per lake with no nulls in precip_mm or pet_mm. The 1992 temperature file covers September to December only. Also, the coordinates listed here are bounding boxes not hydrological catchments, Lake Turkana's bounding box reaches 4.70°N, so the download extent's 6°N boundary sits 1.3° above it, while the Omo headwaters, which supply over 80% of Turkana's inflow, lie north of 6°N and outside the domain entirely. The download product is daily but every analysis in the report is monthly. The reduction between them is performed in stage 05, where potential evaporation is also adjusted to enter as a positive magnitude. ERA5 natively reports it as negative, so stage 05 negates it. The check for this is Turkana, which should come out at 283 mm of precipitation against 2,103 mm of

potential evaporation, a water-balance of -1820 mm/yr, the only negative value among the seven lakes. Leaving ERA5's native sign would give roughly +2386 mm/yr instead.

3.5. Per-lake bounding boxes

Given as (S, N, W, E) in decimal degrees, matching the LAKE_BOXES dictionary in the climate extraction notebook.

Lake	South	North	West	East
Victoria	-3.10	0.6	31.5	34.9
Tanganyika	-8.90	-3.30	29	31.2
Malawi	-14.60	-9.40	33.9	35.3
Turkana	2.4	4.7	35.8	36.8
Albert	1	2.4	30.3	31.5
Edward	-0.70	-0.05	29.2	30
Kivu	-2.60	-1.50	28.8	29.5

TABLE T 5: Per-lake bounding boxes used to reduce the ERA5 grid to one monthly value per lake per variable. Every box lies inside the single regional download extent of Table T 4.

3.6. Derived products, and why they are committed

Satellite altimetry repositories reprocess their archives, and the ERA5 daily product is recomputed on every retrieval, so a download made today may not be identical to the one used originally. Committing the four derived products enables a reproducer to check if they are identical.

File	Reference Copy	Produced by
African_Great_Lakes_Water_Levels.xlsx	Reference Outputs/Lake Level Outputs/	Stages 00-01, collation
Unified_BiasAligned_Levels.xlsx	Reference Outputs/Fusion Outputs/	Stage 02, fusion
Unified_Interpolated_Levels.xlsx	Reference Outputs/Gap Interpolation Outputs/	Stage 03, interpolation
Lake_Climate_Monthly.xlsx	Reference Outputs/Climate Data Extraction Outputs/	Stage 05, ERA5 extraction

TABLE T 6: Committed derived products.

Unified_Interpolated_Levels.xlsx is the file every modelling stage reads, so if a reproducer is interested in the forecasting results rather than the record construction, they can start at stage 06 and skip stages 00-05 entirely. But Stage 06 onward read from Code Outputs/, not from Reference Outputs/, so starting at Stage 06 would require copying the last two files of TABLE T 6 over to the appropriate Code Outputs subfolders.

4. Computational Environment

4.1. Software

The environment is managed with conda, environment.yml in the repository root recreates it exactly; requirements.txt carries the pip pins for anyone not using conda, but the conda specification is the one the reported results were produced under.

Package	Version	Used for
Python	3.13.9 (Anaconda, Clang 20.1.8)	All analysis
numpy	2.3.5	Arrays; the GNAR and GNARX estimators are implemented directly in NumPy
pandas	2.3.3	Series handling, fusion, interpolation
scipy	1.16.3	Hierarchical clustering, statistical tests
statsmodels	0.14.5	ARIMA, SARIMA, SARIMAX, VAR, VARX, ADF, KPSS, STL, Ljung–Box
scikit-learn	1.7.2	Present in the environment, no stage imports it
xarray	2026.4.0	Reading the ERA5 NetCDF grids (conda-forge channel)
netcdf4	1.7.4	NetCDF backend
matplotlib	3.10.6	All figures
seaborn	0.13.3	Heatmaps and seasonal box plots
cartopy	0.25.0	The study area map, stage 22
cdsapi	0.7.7	The CDS API client
openpyxl	3.1.5	Excel input and output
pytorch	2.10.0	Graph convolution and GRU. Build <code>cpu_openblas_py313h355ee32_1</code> – see note below
jupyter	1.1.1	Notebook execution
jupyterlab	4.4.7	Notebook environment

TABLE T 7: Software environment. All packages from the Anaconda default channel except xarray, which came from conda-forge.

The PyTorch build string is part of the specification. This is a CPU-only OpenBLAS build on Apple Silicon: no CUDA, and no Metal acceleration. Linear algebra backend and thread count both influence floating-point summation order, which is the mechanism behind the seed sensitivity described in Section 8.2. A reproducer on a CUDA build should expect the neural results to move even with identical seeds.

4.2. Hardware and threading

Machine	Apple MacBook Air (2020), Apple M1, 8GB unified memory
Operating System	macOS Tahoe 26.3 up to macOS Tahoe 26.6.1
Accelerator	None. CPU-only, OpenBLAS backend
BLAS threads	Pinned to 1 in Stage 10-14, 16 and 17, and in forecast_gnarx.py. The three neural stages, 15, 18 , and 19, do not pin them

TABLE T 8: Hardware

5. The Pipeline

Twenty-three notebooks, executed in numerical order. Each uses the previous stages' outputs, and all of them write to Code Outputs/. The stage number is the notebook number.



Figure 1: The pipeline. Grey arrows show the run order. Every stage writes into Code Outputs/.

Notebook	Principal Outputs
00_collate_victoria_tanganyika	African_Great_Lakes_Water_Levels.xlsx – first two lakes
01_collate_other_five_lakes	African_Great_Lakes_Water_Levels.xlsx – all seven lakes; Albert/Edward/Kivu_DAHITI.xlsx; African_Great_Lakes_Water_Levels_BACKUP.xlsx
02_fusion	Fusion_Offsets_Consistency.csv, Fusion_TransitionJump_Check.csv, Unified_BiasAligned_Levels.xlsx, two diagnostics
03_gap_interpolation	Coverage_Diag.png, Coverage_Table.csv, Unified_Interpolated_Levels.xlsx
04_era5_download	Raw ERA5 NetCDF
05_climate_extraction	Lake_Climate_Monthly.xlsx
06_eda	EDA_01 ... EDA_10
07_climate_eda	CLIM_01 ... CLIM_05
08_climate_delta_leadlag	CLIM_06_delta_leadlag, CLIM_06_delta_lagged
09_climate_stock_flux	CLIM_07_wb_vs_dlevel, CLIM_07_stock_vs_flux, CLIM_07_wb_dlevel_stats
10_ccm_network	CCM_00 ... CCM_13
11_baselines	Baseline_orders, Baseline_metrics, Baseline_predictions, FC_forecast_plots; and compatibility copies FC_orders.csv and FC_metrics.csv into Arima Forecast Outputs
12_var_varx	FC4_metrics
13_varx_diagnostic	VARX_diagnostic_metrics
14_sarimax	FC5_climate_metrics, FC5_exog_coeffs
15_gnn	FC6_gnn_metrics
16_gnarx	FC7_gnarx_metrics, FC7_gnarx_coeffs, FC7_gnarx_orders
17_gnarx_adjacency_null	FC8_adjacency_sensitivity, FC8_null_distribution, FC8_sensitivity_summary
18_gnn_adjacency_null	FC9_gnn_adjacency, FC9_gnn_null, FC9_gnn_adjacency_summary
19_gnn_seed_variance	FC10_gnn_seed_variance, FC10_seed_summary, FC10_seed_paired
20_scaled_metrics	Every *_scaled.csv – the nRMSE and MASE columns
21_report_figures	FIG_01 ... FIG_06, FIG_data_audit
22_study_area_map	Lake Figure.svg, in Study Map Outputs

TABLE T 9: Pipeline stages in execution order.

5.1. Stage Detail

00-01 Source Collation

Each lake's source records are read and joined onto a common monthly index. Output is `African_Great_Lakes_Water_Levels.xlsx`, one column per lake per source. Notebook 00 handles Victoria and Tanganyika, notebook 01 does the other 5 lakes. All five DAHITI series are embedded as literal values in the notebooks, Victoria and Tanganyika in Notebook 00, Albert, Edward and Kivu in 01, so none of them needs downloading. Stage 01 also writes `Albert_DAHITI.xlsx`, `Edward_DAHITI.xlsx`, and `Kivu_DAHITI.xlsx` into Code Outputs/Lake Level Outputs in its first cell and reads them back in its second; they are intra-stage intermediates, not downloads and not inputs. A backup copy of the master file, `African_Great_Lakes_Water_Levels_BACKUP.xlsx`, is written alongside on every run. Malawi and Turkana do not have a DAHITI record.

02 Multi-mission fusion

C3S is adopted as the universal backbone. For each of G-REALM and DAHITI the median difference to C3S over overlapping months is computed and subtracted, placing every source on a common datum, remaining C3s gaps are filled with offset-corrected G-REALM first and DAHITI second.

03 Gap Interpolation

Interior gaps are filled by time-weighted interpolation with `limit_area="inside"`, so no value is extrapolated beyond observed data, and every filled month carries an `is_interpolated` flag. `MAX_GAP` is set to `None`, meaning no cap on run length.

04 ERA5 Retrieval

The CDS API request of Table T 4, issued year by year for temperature and half-year by half-year for precipitation and potential evaporation.

05 Climate Extraction

Xarray opens each file, normalising coordinate names across the several ERA5 conventions and dropping the singleton ensemble dimension. For each lake the grid is masked to the lake's bounding box, averaged over the cells inside it, then resampled to months.

06 Exploratory Analysis

STL decomposition with period 12 and robust=True separates trend, seasonal and remainder. Dependence analysis uses the remainder throughout, because raw levels share a slow multi-decadal trend and a strong annual cycle that would otherwise generate correlation reflecting shared forcing rather than interaction. Stationarity is assessed by ADF and KPSS jointly. Trends use Mann-Kendall with a tie correction and Sen's slope. The Z statistic is not corrected for autocorrelation, which inflates it substantially on monthly level series; Sen's slope is unaffected so trend magnitude and direction are secure where the nominal significance is not. Lead-lag structure uses prewhitened cross-correlation. PCA is a singular value decomposition of the standardised residual matrix over the listwise-common window. Every output of this stage is computed on the full record, not the canonical modelling window.

07-09 Climate lead-lag and stock-flux

Stage 07 correlates each climate variable against level. 08 repeats the analysis against the first-differenced level, deseasonalising both series first, and is the specification the models use: a lake's level is the running integral of its water balance, so comparing instantaneous rainfall against accumulated level is a mismatch of stock-against flux. Stage 09 plots the contrast between the two.

10 CCM network estimation

Convergent Cross Mapping with time delay $\tau=1$ throughout. The embedding dimension is selected per lake by a parsimony rule, the estimator is run across four preprocessing variants, and only edges called by all four are reported as the robust core. Significance is tested against 200 Ebisuzaki phase-randomised surrogates per directed edge; the strict convergence threshold is calibrated separately, against 200 independent AR(1) control pairs, which also give the false-positive rate.

11 Univariate baselines

Four models per lake: random walk, seasonal naïve, ARIMA (p,1,q) and SARIMA (p,1,q)(P,D,Q)12. Orders are selected by AIC grid search on the training window only. SARIMA is the reference which every joint model is scored against.

12-13 VAR, VARX and the diagnostic variants

VAR on differenced levels with eleven monthly dummies as deterministic terms, so the joint model is not handicapped against SARIMA's seasonality. VARX adds the climate block. Forecasts are made in differences and cumulated back to levels. Stage 13 runs the diagnostic variants, the zero-filled, original, DMI-only and oracle exogenous polices, side by side.

14 Per-lake SARIMAX

Each lake's SARIMA order plus only its own covariates: its own water-balance anomaly at lag 1 and DMI at that lake's own lag, taken from stage 08.

15 Graph Neural Network

A graph convolution followed by a gated recurrent unit, in plain PyTorch without torch_geometric. The adjacency is the residual-correlation matrix written by Stage 06, EDA_06_corr_residual.csv, with an identity added on top; that file already carries a unit diagonal, so the self-weight on the diagonal is 2.0 rather than 1.0. Twelve-month input window, full-batch training, three-seed ensemble, early stopping, recursive multi-step forecasting.

16-17 GNAR, GNARX and the linear adjacency null

The Generalised Network Autoregressive model and its exogeneous extension, implemented directly in NumPy in gnarx_core.py as opposed to being taken from a package, with the rolling origin harness on forecast_gnarx.py. Order selection by BIC on the training window. Standard errors are clustered by date, because the seven lakes at one date are contemporaneously correlated and naïve OLS standard error would be too small. Stage 17 substitutes 200 random density-matched graphs for the CCM adjacency, reusing Stage 16's own fitting and rolling functions.

18-19 The neural adjacency null and seed variance

Stage 18 evaluates eight adjacency variants, one of which is an identity matrix representing no network, plus 30 random density-matched graphs. Stage 19 repeats the evaluation of five graphs ten times with disjoint seed triples, changing nothing else.

20 Scale-free metrics

A post processor over every metrics file, RMSE and MAE are in metres and therefore not comparable across lakes whose dynamic ranges differ by a factor of five, so two scale-free comparisons are added. nRMSE as a percentage of the lake's training standard deviation, and MASE against the training seasonal-naïve benchmark at a twelve-month period.

21-22 Report figures and the study area map

Stage 21 produces six figures from the metrics files, five of which are numbered figures in the report, with the only one not appearing being FIG_05_parsimony.svg. Stage 22 produces the study area map.

6. Parameters

The key fixed settings within the analysis, if any of these are changed the results of any reproduction will differ from the reported numbers.

6.1. Canonical Evaluation Protocol

Setting	Value
Lakes	Victoria, Tanganyika, Turkana, Malawi, Albert, Edward and Kivu
Canonical window	1995-06-01 to 2025-12-01 – 367 months, 7 lakes, 0 missing
Window Start	Albert's record start, the latest of the seven
Window End	Two to three months of available data deliberately excluded, giving a clean 60-month test period
Training	1995-06-01 to 2020-12-01 – 307 months
Test	2021-01-01 to 2025-12-01 – final 60 months
Horizons	1, 3, 6, 12 months
Scheme	Rolling origin
Shared baseline	Baseline_predictions.csv, written once by Stage 11
Forecast counts	60/ 58/ 55/ 49 per lake per horizon
Refit – VAR, VARX	Re-estimated at every origin
Refit – SARIMA, SARIMAX, GNARX	Estimated once on the training window, then filtered forward
Scoring	RMSE and MAE in metres; nRMSE against the training standard deviation; MASE against the training seasonal naïve
Interpolated targets	Excluded from scoring in stages 11, 16, and 17. There are no interpolated months inside the test window for any lake
Skill	Percentage against SARIMA, computed from RMSE rounded to four decimal places

TABLE T 10: The canonical evaluation protocol. Shared by every model.

6.2. Scoring denominators

The two scale-free metrics are normalised per lake, from the training portion of the canonical window, 1995-06 to 2020-12, 307 months. Reproducing these seven pairs of numbers before comparing any metric is the easiest way to confirm the scoring is aligned.

Lake	Training SD (m) – the nRMSE denominator	MASE scale (m)
Albert	0.567	0.425
Edward	0.307	0.299
Kivu	0.327	0.292
Malawi	0.644	0.290
Tanganyika	0.469	0.296
Turkana	1.128	0.667
Victoria	0.551	0.317

TABLE T 11: Per-lake scoring denominators, from stage 20. nRMSE is $100 \times \text{RMSE} \div \text{training SD}$; MASE is $\text{MAE} \div \text{the mean absolute twelve-month difference on training}$. MASE below 1 beats seasonal naive.

6.3. Preprocessing and exogenous policy

Setting	Value
Fusion Backbone	C3S, offset computed as the median difference over overlapping months
Gap-fill priority	Offset-corrected G-REALM first, then DAHITI
Interpolation	Time-weighted, limit_area="inside", every filled month flagged
Maximum gap filled	MAX_GAP = None. Every internal gap is filled regardless of run length, so limit_area is the only acting constraint
STL	Period = 12, robust = True
Deseasonalisation modelling stage	Monthly climatology computed on training rows only then applied to the whole series
Deseasonalisation stages 07-09	Calendar-month mean over full record. This is the sample on which the exogenous lead lags are selected, so the selection window includes the test period
Water-balance lag	One month, fixed for all seven lakes
DMI lags, per lake	Victoria 2, Malawi 3, Albert 3, Tanganyika 4, Kivu 5, Edward 6, Turkana 11
DMI lag selection	Lag maximising cross-correlation between deseasonalised DMI and deseasonalised monthly level change
DMI lag ceiling	$\text{np.clip}(\text{lag}, 1, 11)$, in stage 14.
Index choice	DMI over Niño 3.4, DMI attains the larger peak correlation against level change at five of the seven lakes
Future climate, row o	Holds climate observed in month o-1
Future climate, row o+1 onward	The lake-level climate anomaly set to 0, and DMI held at last observed value
Oracle variant	Substitutes actual future climate

TABLE T 12: Preprocessing and the leakage-free exogenous policy.

6.4. Convergent Cross Mapping

Setting	Value
Time delay τ	1 throughout, appropriate for monthly data
Embedding dimension E	Searched over 1-10 per lake per variant
E selection rule	Parsimony: the smallest E whose simplex ρ falls within approximately 0.01 of the maximum attained, not argmax.
Selected E, primary graph	Albert 8, Edward 7, Kivu 8, Malawi 7, Tanganyika 8, Turkana 6, Victoria 10
Preprocessing variants	{STL residuals, first differences} x {full record, training window} – four in total
Robust core	Only edges called by all four variants: 19 edges
Negative control	200 independent AR(1) pairs with lake-like autocorrelation, ϕ drawn uniformly on 0.65-0.90 put through the identical pipeline. 9 edges called, a 4.5% false positive rate against a nominal 5%. Each trial's own significance test uses 200 Ebisuzaki surrogates
Strict convergence threshold	The 95 th percentile of $\Delta\rho$ across the 200 control trials
Adjacency orientation	$A[i,j]>0$ means j influences i
Random seed	SEED = 20260807
Library sizes L	20, 30, 45, 65, 90, 120, 160, 200, 250, and $L_{\max} = n - (E_{\max} - 1)$
Edge rule	Converged ($\Delta\rho > 0$ and Kendall τ of ρ against $L > 0$) and $q < 0.05$ and ρ at $L_{\max} \geq 0.10$
ρ control	RHO_FLOOR = 0.10
Bootstrap replicates	$n_{\text{boot}} = 100$

TABLE T 13: Convergent Cross Mapping settings.

Convergent Cross Mapping is implemented for this project in `ccm_core.py` in NumPy. No external CCM package is used or required and none appears in the environment. Three distinct graphs objects appear in this project. The 19-edge robust core (`CCM_12_robust_core.csv`) is the reported characterisation of the network and is not a model input. The graph the forecasting stages read is the 20-edge top-3 adjacency (`CCM_09_adjacency_top3_diff_train.csv`), which retains each nodes three highest- ρ incoming edges from the first differences / training-window variant, ρ -weighted rather than binarised. The pool it selects from is the edges that pass the significance test alone: the convergence criterion and the $\rho \geq 0.10$ floor are applied to the robust core, not to this object. The neural network reads a third object, the residual-correlation matrix (`EDA_06_corr_residual.csv`).

6.5. Model Specifications

Model	Specification	Lags p	Exogenous	Parameters
ARIMA	Grid over p, q $\in$ 0-3 with d=1 fixed, lowest AIC	-	-	Per-lake
SARIMA	ARIMA order plus a small seasonal grid by AIC. Seasonal terms are forced	-	-	Per-lake
VAR	Differenced levels, 11 monthly dummies, AIC with VAR_MAXLAG = 4	3	11 dummies	231 total, 154 non-seasonal
VARX	As VAR plus the climate block	1	11 dummies + 7 precipitation anomalies + DMI = 19	189 total, 112 non-seasonal
SARIMAX	Each lake's SARIMA order plus its own water balance at lag 1 and its own DMI lag	-	2 per-lake	Per-lake
GNAR	NumPy implementation, order by BIC	1	-	9
GNARX_wb	GNAR plus shared water-balance coefficient	1	Water balance	9
GNARX_wb_dmi	GNARX_wb plus shared DMI coefficient	1	Water balance + DMI	10
GNN	Graph convolution + GRU, plain PyTorch. Full hyperparameters in Table T 16	12-month window	none	Spatial hidden 16, GRU hidden 24

TABLE T 14: Model specifications and parameter counts. VAR and VARX absorb seasonality with 77 dummy coefficients; GNARX removes it in preprocessing with a training-only climatology, which costs no estimated parameters but is not free in information terms, both counts are given for this reason

6.6. The GNARX estimator

GNAR and GNARX were implemented for this project, but they were not taken from a package, so the settings below are the specification.

Setting	Value
Implementation	gnarx_core.py (estimator) and forecast_gnarx.py (rolling-origin), both NumPy
Model type	“standard” – node-specific α , one shared β per lag and neighbour stage, one shared λ per exogenous variable
Neighbour weighting	WEIGHTED = True. Stage-r neighbours of node i are the nodes at shortest-path distance exactly r in the binarised graph; their weights are the corresponding entries of A^r , row normalised, so each stage contributes a p -weighted mean rather than a plain average. MAX_STAGE = 2
Adjacency preparation	Reindexed to the lake order, asserted to cover all seven lakes and to be non-negative, diagonal set to zero, used untransposed
Lag grid	$p \in \{1, 2, 3\}$
Stage grid	Stage vectors of length p with entries 0-2, non-increasing across lags, and at least one neighbour stage at lag 1
Order selection	BIC, on training rows only
Series modelled	First differences, deseasonalised by a monthly climatology of the differences computed on training rows only
Exogenous lags	exog_lags = (1,)
Water-balance scaling	The deseasonalised water balance is divided by its own training-window standard deviation before entering , so the shared λ is not dominated by the unit difference and is reported in scaled units
DMI block	$Z[t,i] = \text{DMI}[t - (\text{lag}_i - 1)]$, so that with the exogenous lag applied on top the value entering the lake i 's equation at time t is $\text{DMI}[t - \text{lag}_i]$. Head values are back filled with the first observation
Estimation sample	Training rows 1 to 306, 2,135 stacked observations at $p = 1$
Standard errors	Clustered by date, with finite-sample scale. Unclustered OLS errors are written alongside for comparison
Forecast reconstruction	Add the seasonal climatology back to the predicted differences, then cumulative-sum onto the last observed level
Threading	BLAS threads pinned to 1 before NumPy is imported.

TABLE T 15: The GNAR and GNARX estimator, as implemented.

The metrics file contains twice as many models as quoted by the report. Stage 16 runs each of GNAR, GNARX_wb and GNARX_wb_dmi twice, once fitted on the training window and filtered forward, and once re-estimated at every origin, and writes both to FC7_gnarx_metrics.csv, the second set are labelled with _refit. The report uses the fit-once

rows throughout. It is also worth noting the water-balance covariate is scaled, so λ is in scaled units.

6.7. Neural network hyperparameters

Setting	Value
Spatial hidden size	16
GRU hidden	24
Learning rate	5×10^{-3}
Weight Decay	1×10^{-4}
Batching	Full batch
Seed per run	3
Maximum epochs	400
Early-stopping patience	30
Dropout	0.2
Input window	12 months
Multi-step	Recursive

TABLE T 16: Graph convolution and GRU hyperparameters. The hidden sizes are deliberately small: with seven nodes and roughly 360 months, capacity is the binding constraint, and the report treats the resulting instability as a finding.

6.8. Null controls and seed protocol

Control	Design
Linear adjacency null	200 random directed graphs matched on the primary graph's edges, with weights drawn uniformly between the smallest and largest ρ . Seed 20260808. Substituted into GNAR and GNARX_wb only; at the primary order; GNARX_wb_dmi is fitted on the real graphs but not the null
Nonlinear adjacency null	30 random directed graphs, density matched to the CCM top-3 graph and row-normalised, seed 20260808 substituted into the GNN. The identity adjacency is one of the eight real graphs, not part of the null
GNN seed ensemble	Three seeds per run
Single-run seed triple	(0, 1, 2) used for the reported GNN results and the nonlinear null
Seed replication	Ten disjoint triples: (0,1,2), (3,4,5)...(27,28,29) changing nothing else
Point estimates	The ten-replicate means from Stage 19

TABLE T 17: Null controls and the seed protocol.

6.9. Graph clarifications

Graph	Construction	Edges	Shares with primary
ccm_diff_train	The primary graph, CCM_09_adjacency_top3_diff_train.csv	20	-
ccm_stl_train	CCM_09_adjacency_top3_stl_train.csv	21	11/20
ccm_strict_diff	CCM_11_adjacency_strict_diff_train.csv	9	6/20
corr_top3	Three largest residual correlations per row of EDA_06_corr_residual.csv	21	11/20
binary_diff	The primary graph, weights set to 1	20	20/20
complete	Every off-diagonal pair, weights 1	42	20/20

TABLE T 18: The six graphs of the linear null.

Graph	Construction	Normalisation
identity	np.eye(N) – no network	-
corr_abs_asbuilt	residual correlation , entries < 0.2 zeroed. The native 1.0 diagonal is retained and an identity matrix is added, yielding a self-weight of 2.0.	Sym
corr_abs_sym	As above, but the native diagonal is explicitly cleared before adding the identity matrix, testing a node self-weight of 1.0	Sym
corr_abs_row	As corr_abs_sym	Row
corr_signed	Residual correlation with the sign retained, entries < 0.2 zeroed	Sym
ccm_directed	The CCM top-3 graph as-is	row
ccm_symmetric	np.maximum(CCM, CCM.T)	sym
complete	Every off-diagonal pair weight 1	Row

TABLE T 19: The eight graphs of the neural null.

The 30-graph null is row-normalised throughout, while three of the eight real graphs are symmetrically normalised. On a graph this sparse the normaliser is worth far more than the topology: holding the graph fixed and changing only the normaliser moves one-month skill by 17.62 ± 2.91 points, while removing the edge direction under a fixed normaliser moves it by only 1.99 ± 0.73 . The symmetrisation rule is specifically `np.maximum(CCM, CCM.T)`; a different rule gives a different answer. Finally, comparing `corr_abs_sym` to `corr_abs_asbuilt` explicitly isolates the effect of the 2.0 self-weight: reducing the self-weight to 1.0 degrades one-month skill from -15.1% to -26.1%, demonstrating that the network heavily relies on prioritising a lake's own history.

7. Provenance Map

Every numbered figure and table in the report, the stage that produces it and the file that holds the reference copy. Every reference copy sits in the subfolder of Reference Outputs named for its stage, mirroring the layout in Code Outputs. Every output filename is unique across the pipeline.

7.1. Numbered Figures and Tables

Report Item	Stage	Reference file
Figure 1 – study area	22	Lakes Figure.svg, in Study Map Outputs
Figure 2 – The seven fused water-level series	06	EDA_01_levels_overview.svg
Figure 3 – PCA of the seven lakes	06	EDA_08_pca.svg
Figure 4 – Mean forecast skill, all models and horizons	21	FIG_01_master_skill.svg
Figure 5 – Per-lake forecast skill at one month	21	FIG_02_lake_model_heatmap.svg
Figure 6 – Attenuation of β as climate enters	21	FIG_03_gnarx_beta_attenuation.svg
Figure 7 – The calibrated null controls	21	FIG_04_null_controls.svg
Figure 8 – Neural skill across ten seed triples by graph	21	FIG_06_gnn_seed_variance.svg
Table 1 – The fused series	06 and 03	EDA_01_summary_stats.csv for the descriptive columns; Coverage_Table.csv for interpolation percentage
Table 2 – Model dimensions and parameter counts	-	Specification – Table T 14
Table 3 – Degree structure of the 19-edge robust core	10	CCM_13_degree.csv
Table 4 – Mean forecast skill against SARIMA	12, 14, 16, 19 and 20	FC4_metrics, FC5_climate_metrics, FC7_gnarx_metrics and their scaled comparisons. The GNN row is the ten-replicate mean from FC10_seed_summary.csv
Table A1 – Source products	-	Compiled by hand – Table T 1 above
Table A2 – Per-lake altimetry targets and missions	-	Compiled by hand – Table T 2 above
Table A3 – Median offset of each source against C3S	02	Fusion_Offsets_Consistency.csv
Table A4 – Per-lake bounding boxes	-	Specification – the LAKE_BOXES dictionary in Stage 05, reproduced in Table T5 above
Table B1 – Selected embedding dimension per lake	10	CCM_05_E_selection.csv
Table C1 – Normalised RMSE and MASE by model and horizon	20	The matching *_scaled.csv files
Table C2 – Per-lake scoring denominators	20	Reproduced as Table T11 above
Table D1 – GNARX coefficients with date-clustered SEs	16	FC7_gnarx_coeffs.csv
Table D2 – Order selection	16	FC7_gnarx_orders.csv

TABLE T 20: Report floats and the files that reproduce them. Figures 1 to 8 and Tables 1 to 4 are in the main text; Tables A1 to D2 are in the report's own appendices.

8. Known deviations

This section is here to clarify key reasons why a reproducer's numbers may differ from the reported ones.

8.1. Two sampling windows appear in the reported results

Every exploratory output, descriptive statistics, stationarity tests, trend tests, correlation matrices, the PCA, is computed on the full record, 370 to 403 months per lake running to 2026-03. Every model result is computed on the 367-month canonical window of Table T 10. The PCA is listwise over 369 common months, while the correlation matrices are pairwise complete, so each cell uses a different number of months. This means that the ACF and PACF that motivated the ARIMA search ranges were computed on the full record including the test period. The orders themselves were selected by AIC on the training window, so the leak is into the candidate ranges only. This is a mild leak, it is a limitation of the design rather than a defect, and it is reproduced faithfully by running the stages in order.

8.2. The neural results are seed-dependent

The single-run figures use seed triple (0, 1, 2). That triple ranks the first of ten at three of four horizons, so the single-run result is the best replicate not a typical one; the ten-replicate means are what the report uses as point estimates. Reproducing the single-run figures requires the same seeds, the same PyTorch build and the same thread configuration. Reproducing the means requires the ten disjoint triples in Table T 17.

8.3. The ERA5 product is daily and the analysis is monthly

The reanalysis product used is the post-processed daily statistics, not a monthly-means product, and the CDS computes it at retrieval time rather than serving it from an archive. This has two main consequences; first this means the DOI alone does not identify the data; the daily statistic, the sampling frequency and the time zone are all chosen in the

request, and any change in request would change the received values, hence why the request in Table T 4 must be reproduced exactly. Second, an aggregation from daily to monthly takes place between the download and every reported climate figure in Notebook 05.

8.4. KPSS p-values are clipped

Statsmodels interpolates KPSS p-values from a table and clips them to the interval [0.01, 0.10]. Every 0.01 in the stationarity outputs means “at most 0.01” and every 0.1 means “at least 0.10”. Values strictly inside the interval, such as Tanganyika’s 0.0442, are genuine, interpolated values. Boundary cases should therefore be read and quoted as inequalities rather than as exact numbers.

8.5. The DMI lead-lag is data-dependent

The seven DMI lead lags are not hard coded in stage 14, they are read from CLIM_06_delta_leadlag.csv, produced by stage 08 from the committed dmi.csv. This has one key consequence, because NOAA recomputes the DMI whenever its underlying sea-surface temperature dataset is revised, re-downloading the index later could result in different results. As a result of this, dmi.csv is committed to the repository with its access date, and a reproducer working from the committed file obtains the reported numbers exactly.